

Curriculum Vita

Clark Evans, Ph.D.

Personal Information

Current Address

NOAA/OAR/Global Systems Laboratory
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Boulder, CO 80305-3328

Last Updated: 24 April 2026

Contact Information

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<https://gsl.noaa.gov/about/profiles/clark-evans>

Education

2009	Ph.D., Florida State University, Meteorology
2006	M.S., Florida State University, Meteorology
2004	B.S. (Magna Cum Laude), Florida State Univ., Meteorology (Minors: Physics, Math)

Experience

2024-present	Research Physical Scientist and Branch Chief, NOAA/OAR/GSL, Boulder, CO
2021-2024	Professor, UW-Milwaukee, Milwaukee, WI
2021-2024	Chair, Atmospheric Science Program, UW-Milwaukee, Milwaukee, WI
2016-2021	Associate Professor, UW-Milwaukee, Milwaukee, WI
2014-2020	Chair, Atmospheric Science Program, UW-Milwaukee, Milwaukee, WI
2011-2016	Assistant Professor, UW-Milwaukee, Milwaukee, WI
2009-2011	Postdoctoral Fellow, UCAR/Advanced Study Program, Boulder, CO
2004	Research Assistant, FSU/Florida Climate Center, Tallahassee, FL
2003-2004	Undergraduate Research Assistant, Florida State Univ., Tallahassee, FL

Affiliate/Adjunct Scientific Positions

2024-present	Adjunct Professor, School of Freshwater Sciences, UW-Milwaukee, Milwaukee WI
2019-2024	Affiliate Faculty, Northwestern Mutual Data Science Institute, Milwaukee, WI
2018	Visiting Scientist, NOAA/NWS/Storm Prediction Center, Norman, OK
2013	Visiting Scientist, NCAR/Mesoscale and Microscale Meteorology Lab, Boulder, CO
2012	Visiting Scientist, NOAA/NWS/National Hurricane Center, Miami, FL

Awards and Honors

2023	Faculty Distinguished Public Service Award, UW-Milwaukee
2021	Office of Research/UWM Foundation Research Award, UW-Milwaukee
2021	Faculty Distinguished University Service Award, UW-Milwaukee
2018	Editors' Award, <i>Monthly Weather Review</i> and <i>Weather and Forecasting</i>
2018	Invited Participant, Inaugural AMS Early Career Leadership Academy
2009	Ph.D. Poster Competition Winner, American Meteorological Society 23 rd Conf. on Weather Analysis and Forecasting/19 th Conf. on Numerical Weather Prediction
2004	Recipient, American Meteorological Society Father James B. Macelwane Undergraduate Research Award

2004 **Recipient**, American Meteorological Society/Industry/Government Graduate Fellowship (Sponsored by the Office of Naval Research)

Publications (*italicized* = advised student)

A citation listing is available on my [Google Scholar](#) page. H-index: 14, i10-index: 20.

DeYoung, C. P., and **C. Evans**, 2026: [Observations and High-Resolution Rapid Refresh model verification of the Lake Michigan warm-season marine atmospheric boundary layer](#). *Wea. Forecasting*, in press.

Evans, C., C. R. Holt, and M. B. Smith, 2026: Performance of a real-time North Atlantic tropical cyclone prediction system using the Model for Prediction Across Scales during the 2024 NOAA Hurricane Forecast Improvement Program Real-Time Experiment. *Wea. Forecasting*, in review.

Evans, C., and K. M. Wood, 2026: [Weather phenomena: extratropical transitions](#). *Encyclopedia of Atmospheric Sciences (3rd Ed.)*, W. A. Robinson, Ed., Elsevier, **1**, 587–600.

Olson, J. B., and Coauthors, 2026: [A description of the MYNN-EDMF turbulence scheme](#). NOAA Tech. Memo. OAR GSL-77, 60 pp.

Blount, D. V., **C. Evans**, I. L. Jirak, A. R. Dean, and S. Kravtsov, 2023: [An objective method for clustering observed vertical thermodynamic profiles by synoptic meteorological conditions](#). *Wea. Forecasting*, **38**, 1143–1156.

Kaminski, A. N., and coauthors, 2023: [A 30-year climatology of northeastern United States atmospheric rivers](#). *J. Appl. Meteor. Climatol.*, **62**, 31–40.

Prince, K. C., and **C. Evans**, 2022: [Convectively generated negative potential vorticity enhancing the jet stream through an inverse energy cascade during the extratropical transition of Hurricane Irma](#). *J. Atmos. Sci.*, **79**, 2901–2918.

Sarro, G. M., and **C. Evans**, 2022: [An updated investigation of post-transformation intensity, structural, and duration extremes for extratropically transitioning North Atlantic tropical cyclones](#). *Mon. Wea. Rev.*, **150**, 2911–2933.

Schultz, D. M., and Coauthors, 2022: [How to be a more effective author](#). *Mon. Wea. Rev.*, **150**, 2819–2828.

Prince, K. C., and **C. Evans**, 2020: [A climatology of indirect tropical cyclone interactions in the North Atlantic and western North Pacific basins](#). *Mon. Wea. Rev.*, **148**, 4035–4059.

Schaffer, J. D., P. J. Roebber, and **C. Evans**, 2020: [Development and evaluation of an evolutionary programming-based tropical cyclone intensity model](#). *Mon. Wea. Rev.*, **148**, 1951–1970.

Schultz, D. M., and Coauthors, 2020: [Data availability principles and practice](#). *Mon. Wea. Rev.*, **148**, 4701–4702.

Evans, C., S. J. Weiss, I. L. Jirak, A. R. Dean, and *D. S. Nevius*, 2018: [An evaluation of paired regional/convection-allowing forecast vertical thermodynamic profiles in warm-season, thunderstorm-supporting environments](#). *Wea. Forecasting*, **33**, 1547–1566.

- Nevius, D. S., and **C. Evans**, 2018: [The influence of vertical advection discretization in the WRF-ARW model on capping inversion representation in warm-season, thunderstorm-supporting environments](#). *Wea. Forecasting*, **33**, 1639–1660.
- Prince, K. C., and **C. Evans**, 2018: [A climatology of extreme South American Andean cold surges](#). *J. Appl. Meteor. Climatol.*, **57**, 2297–2315.
- Burlingame, B. M., **C. Evans**, and P. J. Roebber, 2017: [The influence of PBL parameterization on the practical predictability of convection initiation during the Mesoscale Predictability Experiment \(MPEX\)](#). *Wea. Forecasting*, **32**, 1161–1183.
- Evans, C.**, and Coauthors: 2017: [The extratropical transition of tropical cyclones. Part I: cyclone evolution and direct impacts](#). *Mon. Wea. Rev.*, **145**, 4317–4344.
- Grunzke, C. T., and **C. Evans**, 2017: [Predictability and dynamics of warm-core mesoscale vortex formation with the 8 May 2009 “super derecho” event](#). *Mon. Wea. Rev.*, **145**, 811–832.
- Keclik, A. M., **C. Evans**, P. J. Roebber, and G. S. Romine, 2017: [The influence of assimilated upstream, pre-convective dropsonde observations on ensemble forecasts of convection initiation during the Mesoscale Predictability Experiment](#). *Mon. Wea. Rev.*, **145**, 4747–4770.
- Karloski, J. M., and **C. Evans**, 2016: [Seasonal influences upon and long-term trends in the length of the Atlantic hurricane season](#). *J. Climate*, **29**, 273–292.
- Manion, A., **C. Evans**, T. L. Olander, C. S. Velden, and L. D. Grasso, 2015: [An evaluation of Advanced Dvorak Technique-derived tropical cyclone intensity estimates during extratropical transition using synthetic satellite imagery](#). *Wea. Forecasting*, **30**, 984–1009.
- Weisman, M. L., and Coauthors, 2015: [The Mesoscale Predictability Experiment \(MPEX\)](#). *Bull. Amer. Meteor. Soc.*, **96**, 2127–2149.
- Burghardt, B., **C. Evans**, and P. Roebber, 2014: [Assessing the predictability of convection initiation across the High Plains using an object-based approach](#). *Wea. Forecasting*, **29**, 403–418.
- Evans, C.**, D. F. Van Dyke, and T. Lericos, 2014: [How do forecasters utilize output from a convection-permitting ensemble forecast system? Case study of a high-impact precipitation event](#). *Wea. Forecasting*, **29**, 466–486.
- Evans, C.**, M. L. Weisman, and L. F. Bosart, 2014: [Development of an intense, warm-core mesoscale vortex associated with the 8 May 2009 “super derecho” convective event](#). *J. Atmos. Sci.*, **71**, 1218–1240.
- Weisman, M. L., **C. Evans**, and L. F. Bosart, 2013: [The 8 May 2009 “super derecho”: analysis of a realtime explicit convective forecast](#). *Wea. Forecasting*, **28**, 863–892.
- Evans, C.**, and Coauthors, 2012: [The PRE-Depression Investigation of Cloud-systems in the Tropics \(PREDICT\) field campaign: perspectives of early career scientists](#). *Bull. Amer. Meteor. Soc.*, **93**, 173–187.
- Evans, C.**, R. S. Schumacher, and T. J. Galarneau, Jr., 2011: [Sensitivity in the overland reintensification of Tropical Cyclone Erin \(2007\) to near-surface soil moisture characteristics](#). *Mon. Wea. Rev.*, **139**, 3848–3870.

Evans, C., and R. E. Hart, 2008: [Analysis of the wind field evolution associated with the extratropical transition of Bonnie \(1998\)](#). *Mon. Wea. Rev.*, **136**, 2047–2065.

Hart, R. E., J. L. Evans, and C. Evans, 2006: [Synoptic composites of the extratropical transition lifecycle of North Atlantic tropical cyclones: factors determining post-transition evolution](#). *Mon. Wea. Rev.*, **134**, 553–578.

Funded Grants and Contracts

I am the sole PI on all funded grants and contracts except where listed.

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|------------------|--|
| 2026-2028 | National Oceanic and Atmospheric Administration
"FY26 Precipitation Prediction Grand Challenge Supplemental: Improving Precipitation Forecasts by Refining the Representation of Multi-Scale Moist Physical Processes." \$797,155; 5/1/26-4/30/28. |
| 2026-2028 | National Oceanic and Atmospheric Administration
"FY26 Precipitation Prediction Grand Challenge Supplemental: Leveraging TPOS/TEPEX-E Observations to Advance Upper-Ocean Mixing and Lower-Atmosphere Turbulence and Cloud Representations and Improve UFS Forecasts." \$425,149; 5/1/26-4/30/28. |
| 2025-2028 | National Oceanic and Atmospheric Administration
"Internal JTTL: Enhancing the Unified Gravity Wave Physics Suite for Improved Scale-Awareness and Middle-Atmosphere Prediction in Global and Regional UFS Applications." \$697,733 (\$397,733 to NOAA/OAR/GSL); 7/1/25-6/30/28. Lead PI; co-PIs: F. Yang (NOAA/NWS/EMC), M. Toy (NOAA/OAR/GSL and CU/CIRES). |
| 2025-2026 | Federal Aviation Administration
"FY25-26 GSL Model Development and Enhancement." \$675,000 (\$220,000 to the Physics Branch led by C. Evans); 5/1/25-4/30/26. Co-PI; lead PI: T. Ladwig (GSL). |
| 2024 | National Science Foundation
"AGS-FIRP Track 1: Learning by Doing: Observing the Lake Michigan Lake-Breeze Circulation." AGS-2347093; \$47,200 (\$22,270 to C. Evans); 3/1/24-8/31/24. |
| 2023-2024 | UWM Discovery and Innovation Grant
"Downstream Impacts of Extratropical Transition of Tropical Cyclones in a Changing Climate." \$112,717; 7/1/23-12/31/24. Co-PI; lead PI: S. Kravtsov (UWM). |
| 2023 | Hamilton Family Foundation
"Observing the Lake Michigan Marine Atmosphere Boundary Layer." \$4,500; 1/1/23-12/31/23. |
| 2022-2026 | U.S. Department of Energy
"Establishing a Holistic Understanding of the Circulations of Mesoscale Convective System Stratiform Regions." Award #DE-SC0023057; \$716,831 (\$276,265 to C. Evans); 8/1/22-3/31/26. Co-PI; lead PI: R. Adams-Selin (AER). |

2021-2023	National Science Foundation "CC* Compute: A Balanced Cluster for Science and Engineering in the Great Lakes Region." OAC-2126229; \$400,000; 10/1/21-9/30/23. Co-PI; lead PI: P. Chang (UWM).
2021-2022	Unidata Equipment Program "Upgrading THREDDS and Deploying JupyterHub at the University of Wisconsin-Milwaukee to Support Education and Research." \$10,672; 6/1/21-5/31/22.
2019-2022	National Science Foundation "Thermodynamics of Tropical Cyclone Overland Maintenance and Intensification." AGS-1911671; \$408,577; 6/1/19-5/31/22.
2019-2021	National Oceanic and Atmospheric Administration "VORTEX-SE: Quantifying the Influence of Sea-Surface Temperature Uncertainty on Cool-Season Severe Weather Events." NA19OAR4590208; \$203,527; 9/1/19-8/31/21.
2018-2020	National Oceanic and Atmospheric Administration "Round 3 of R2O Initiative – NOAA Testbeds: Evaluation of GFS-FV3 Vertical Profile and Thermodynamic Environment Fidelity." NA18NWS4680062; \$210,369 (\$190,369 to C. Evans); 9/1/18-8/31/20. Lead PI; co-PI: I. L. Jirak (NOAA/NWS/SPC).
2018-2019	UWM Research Growth Initiative "A Climatology of Indirect Tropical Cyclone Interactions." \$55,243; 7/2/18-7/1/19.
2017-2019	National Oceanic and Atmospheric Administration "FY 2017 Joint Hurricane Testbed: Evolutionary programming for probabilistic tropical cyclone intensity forecasts." NA17OAR4590137; \$199,527; 7/1/17-6/30/19. Co-PI; lead PI: P. Roebber (UWM).
2015-2018	National Science Foundation "Collaborative Research: SI2-SSI: Big Weather Web: A common and sustainable big data infrastructure in support of weather prediction research and education in universities." ACI-1450439; \$2,000,000 (\$164,381 to C. Evans); 8/1/15-7/31/18. Co-PI; lead PI: C. Maltzhan (UC-Santa Cruz).
2015-2016	Unidata Equipment Program "Deployment of AWIPS-II at the University of Wisconsin-Milwaukee." \$11,908, 6/1/15-5/31/16.
2014-2017	National Science Foundation "Numerical Assessment of the Practical and Intrinsic Predictability of Warm-Season Convection Initiation Using Mesoscale Predictability Experiment (MPEX) Data." AGS-1347545; \$456,206; 6/1/14-5/31/17. Lead PI; co-PI: P. Roebber (UWM).
2012-2013	UWM Graduate School Research Committee "An Assessment of Thunderstorm Development Forecast Successes and Failures from Very High Resolution Numerical Weather Forecasts." \$12,611; 7/1/12-6/30/13.

- 2012-2013 Unidata Equipment Program**
 "Installation of RAMADDA, THREDDS, and LDM at UWM." \$7,177; 6/1/12-5/31/13.
 Co-PI; lead PI: P. Roebber (UWM).
- 2011-2012 COMET Partners Program**
 "Extreme Precipitation Across the Tallahassee, FL NWS Forecast Area Associated
 with Tropical Storm Fay (2008): Physical Understanding and Ensemble-Based
 Forecast Utility." \$9,990; 7/13/11-8/31/12. Lead PI; co-PI: D. Van Dyke
 (NOAA/NWS).

Teaching Experience *(* = newly developed course; ^ = developed own materials)*

- Tropical Meteorology** (Atm Sci 470*)
 Spring 2024, Spring 2022, Spring 2020, Spring 2018, Spring 2016, Spring 2014, Spring 2012
- Numerical Weather Prediction** (Atm Sci 730*)
 Fall 2023, Fall 2021, Fall 2019, Fall 2017, Fall 2015, Fall 2012
- Introductory Atmospheric Science Seminar** (Atm Sci 101*)
 Fall 2023, Fall 2022
- Synoptic Meteorology II** (Atm Sci 361^)
 Spring 2023, Spring 2019, Spring 2015, Spring 2013
- Synoptic Meteorology I** (Atm Sci 360^)
 Fall 2022, Fall 2018, Fall 2014
- Mesoscale Meteorology** (Atm Sci 460^)
 Spring 2017
- First-Year Seminar: Probability, Uncertainty, and Communication** (Atm Sci 194*)
 Fall 2016
- Survey of Meteorology** (Atm Sci 100^)
 Spring 2014, Fall 2013
- Current Weather Discussion** (MET 3520^, Florida State University)
 Spring 2008

Student Advisees

Graduate Advisees

- 2023-2025 Kathryn Boyle**
 M.S. 2025; co-advised by S. Kravtsov; now incoming Ph.D. student, North Carolina
 State Univ., Raleigh, NC
- 2022-2024 Collin DeYoung**
 M.S., 2024; now Instructor of Meteorology, Central Michigan Univ., Mt. Pleasant, MI
- 2021-2024 Ariel Tickner-Ernst**
 M.S., 2024; now Mitigation & Recovery Coordinator, Cumberland County, ME

2020-2021	Michelle Storm (née: Spencer) M.S., 2021; Ph.D., 2025 at the Univ. of Oklahoma; now UCAR Advanced Study Program Postdoctoral Fellow, Boulder, CO
2019-2025	Dillon Blount M.S., 2021; Ph.D. 2025; now Asst. Professor (tenure-track) at Ohio Univ., Athens, OH
2019-2024	Michael Vossen M.S., 2021; Ph.D. (ABD) at time of departure; now Senior Project Professional with the Antea Group USA, St. Paul, MN
2017-2019	Jesse Schaffer M.S., 2019; completed M.Ed. in 2022 at George Mason Univ.
2016-2022	Kevin Prince M.S., 2018, Ph.D., 2022; now NRC Post-Doctoral Fellow, NRL, Monterey, CA
2016-2018	Aidan Kuroski M.S., 2018; now Meteorologist with NWS, Milwaukee/Sullivan, WI
2016-2018	David Nevius M.S., 2019; now Flight Superintendent with Delta Airlines, Atlanta, GA
2015-2017	Caitlin Cook (née: Crossett) M.S., 2017; Ph.D., 2022 at the Univ. of Vermont; now Asst. Professor (tenure-track) at UNC-Asheville, Asheville, NC
2014-2016	Alexandra Kelly (née: Keclik) M.S., 2016; now NWS Central Region IDSS/WCM Program Mgr., Kansas City, MO
2014-2016	Bryan Burlingame M.S., 2016; now Chief Technology Officer with SparkPlug, Milwaukee, WI
2014-2016	Caleb Grunzke M.S., 2016; now Meteorologist with NWS, Twin Cities/Chanhassen, MN
2013-2015	Juliana Karloski M.S., 2015; now Educational Instructor with Space Center Houston, Houston, TX
2012-2014	Alex Manion M.S., 2014; now Meteorologist with NWS, Detroit/Pontiac, MI
2011-2013	Brock Burghardt M.S., 2013; Ph.D., 2017 at the Texas Tech Univ.; now Certified Consulting Meteorologist in private practice

UWM Undergraduate Student Advisees

2023-present	Kade Barkas	
2022-2024	Drew Hickok	
2018-2021	Anna Prince (née: Kaminski)	(2021 AMS Father James B. Macelwane Awardee)
2018-2020	Giorgio Sarro	(2020 AMS Father James B. Macelwane Awardee)
2018	Marie Freres	

NOAA/NCAR Student Advisees

2026	Matt Oehler	(William M. Lapenta NOAA Student Intern Program)
2025	Darri Stuber	(William M. Lapenta NOAA Student Intern Program)
2010	Dereka Carroll-Smith	(NSF NCAR SOARS)

Ph.D. Dissertation/M.S. Thesis Committee Member (all at UW-Milwaukee excepted where listed)

Dissertations: Andrew Westgate (2024), Ilijana Mastilovic (2023), Tim Thielke (2022), Austin Harris (2022), Brian Griffin (2016), Noriyuki Sugiyama (2015), Dawn Kopacz (2015)

Theses: Ander Ortiz (2025, Univ. of Arizona), Alex Moxon (2025), Skylar Gertonson (2025), Victoria Lang (2022), James Ryan (2020), Teresa Turner (2020), Andrew Westgate (2020), Christian Grimm (2018), Andrea Honor (2018), Cory Rothstein (2018), Tim Thielke (2018), Lily Chapman (2017), Russell Danielson (2017), Austin Harris (2016), Kaitlyn Heinlein (2016), Timm Uhlmann (2016), Justin Weber (2015), Josh Verbeten (2014), Joseph Pehoski (2013), Jeremy Duggan (2012), John Peters (2012), Marc Pilon (2012), Zach Uttech (2012)

Undergraduate Capstone Supervision

Drew Hickok (2024), Kyle Zur (2022), Anna Kaminski (2021), Giorgio Sarro (2020), Ashley Schils (2020), Devon Bernick (2019), Austin Scheib (2018), Mackenzie Nuthals (2017), Alec Muniz (2016), Lily Chapman (2015), Kyle Koval (2013), Karleisa Rogacheski (2013), Charles Smith (2013)

Professional Service

National/International Service

2026-present	Member , AMS Ad Hoc Community Modeling Committee
2025-present	Member , AMS Planning Commission
2025-present	Member , AMS Awards Oversight Committee
2025-present	Member , Committee on AMS Positions and Statements
2025-present	Member , AMS Annual Meeting Oversight Committee (<i>ex officio</i>)
2025	Panelist , UCAR Early Career Workshop
2024	Panelist , 23 rd Annual AMS Student Conference
2023-present	Commissioner , AMS Scientific and Technological Activities Commission
2023	Chair , Developmental Testbed Center Science Advisory Board
2023	Panelist , AMS Town Hall on Open Science Expectations for Model-Based Research
2022-2023	Co-Chair , AMS Future of Meetings Task Force
2022-2023	Member , AMS Future of Meetings Task Force
2021-2023	Chair , AMS Committee on Weather Analysis and Forecasting
2021	Chair , AMS Weather Analysis and Forecasting Statement Revision Team
2021	Member , NCEP Strategic Planning Team
2021	Member , AMS 102 nd Annual Meeting Health and Safety Task Force
2021	Panelist , 20 th Annual AMS Student Conference
2020-2022	Chair , AMS Annual Meeting Oversight Committee
2020-2023	Member , UCAR Membership Committee
2020-present	Member , Developmental Testbed Center Science Advisory Board
2020	Panelist , 8 th Annual AMS Conference for Early Career Professionals
2019-present	Editor , <i>Monthly Weather Review</i>
2019-2022	Member , AMS Annual Meeting Oversight Committee
2018-2021	Vice Chair , AMS Committee on Weather Analysis and Forecasting
2018	Rapporteur , 9 th WMO International Workshop on Tropical Cyclones
2018	Organizer , AMS Special Symposium on Impact-Based Decision Support Services
2017	Member , AMS 28 th Conf. on WAF/24 th Conf. on NWP Program Committee
2016-2023	Member , AMS Committee on Weather Analysis and Forecasting
2016	Member , AMS Max Eaton Award Selection Committee

2015	Panelist , 14 th Annual AMS Student Conference
2015	Member , 17 th Cyclone Workshop Science Committee
2014	Member , 8 th WMO International Workshop on Tropical Cyclones Working Group
2013-2015	Member , AMS Weather Analysis and Forecasting Statement Revision Team
2013	Panelist , 1 st Annual AMS Conference for Early Career Professionals
2012-2018	Associate Editor , <i>Monthly Weather Review</i>
2012	Member , AMS Max Eaton Award Selection Committee
2012	Rapporteur , 4 th WMO International Workshop on Extratropical Transition
2010	Member , 7 th WMO International Workshop on Tropical Cyclones Working Group
2010	Member , AMS 25 th Conf. on Severe Local Storms Program Committee
2010	Member , AMS 29 th Conf. on Hurricanes/Tropical Meteor. Program Committee

Institutional Service

2025-present	Chair , NOAA/OAR/GSL Awards Committee
2024-present	Member , NOAA/OAR/GSL Science and Technology Council
2024-present	Mentor , CIRES Mentoring Program
2024-2025	Member , NOAA/OAR/GSL Modeling Summit Leadership Team
2023	Lecturer , UWM Admitted Student Days Mock Lecture Series
2022-2023	Vice Chair , UWM Freshwater Sciences Academic Program & Curr. Committee
2022-2023	Member , UWM Freshwater Sciences Dean Search & Screen Committee
2021-2024	Member , UWM Freshwater Sciences Academic Program & Curr. Committee
2021-2024	Faculty Advisor , The Climate Consensus at UWM
2021-2022	Member , UWM Freshwater Sciences Climate/Water Asst. Prof. Search Committee
2020-2024	Member , UWM Research Computing Steering Group
2020-2021	Member , 2030 Implementation Team Undergraduate Experience Working Group
2018-2020	Member , UWM Information Technology Policy Committee
2017-2019	Recruitment Ambassador , UWM College of Letters and Science
2017-2019	Member , UWM Mathematical Sciences Strategic Planning Committee
2017-2018	Member , UWM Mathematical Sciences Undergraduate Committee
2017-2018	Member , UWM Mathematical Sciences Department Manager Search Committee
2017-2018	Member , UWM Mathematical Sciences Merit Committee
2017-2018	Chair , UWM Mathematical Sciences Visiting Assistant Professor Search Committee
2016-2017	Member , UWM Mathematical Sciences Assessment Committee
2014-2023	Coordinator , UWM StormReady University Initiative
2014-2020	Member , UWM Mathematical Sciences Graduate Committee
2013-2014	Chair , UWM Mathematical Sciences Event Organizing Committee
2012-2024	UCAR Member Representative , UW-Milwaukee
2011-2024	Local Manager , WxChallenge Forecasting Competition
2011-2024	Faculty Co-Advisor , UWM Atmospheric Science Club
2011-2016	Member , UWM Mathematical Sciences Colloquium Committee
2011-2016	Member , UWM Mathematical Sciences Event Organizing Committee
2010-2011	Organizer , UCAR/NCAR/MMM 'Dynamics Happy Hour' Seminar Series
2009-2011	Member , UCAR/NCAR/ASP Seminar Organizing Committee

Service to Other Universities

2024	Reviewer , Univ. of North Carolina-Charlotte – Promotion to Professor
2024	Reviewer , Northern Illinois University – Promotion to Professor
2023-2024	Chair and Member , Western Kentucky Univ. Meteorology B.S. Review Committee
2023	Reviewer , University of Kansas – Tenure and Promotion to Associate Professor
2022	Reviewer , Hobart & Wm. Smith Colleges – Promotion to Professor
2019	Reviewer , Northern Illinois University – Tenure and Promotion to Associate Professor
2018	Reviewer , Texas Tech University – Tenure and Promotion to Associate Professor
2016	Reviewer , Hobart & Wm. Smith Colleges – Tenure & Promotion to Assoc. Professor

Community Service

2025-present **Member**, State of Colorado Natural Disaster Mitigation Enterprise Board
2025-present **Member**, High Plains Library District Friends & Foundation Board of Directors
2024 **Member**, Village of Grafton, WI Finance and Personnel Committee
2023-2024 **Trustee**, Village of Grafton, WI (elected position)
2023-2024 **Member**, Village of Grafton, WI Board of Public Works
2022-2023 **Vice President**, U.S.S. Liberty Memorial Public Library (Grafton, WI) Library Board
2018-2020 **Participant**, ESWN Science-a-Thon #dayofscience
2016-2023 **Trustee**, U.S.S. Liberty Memorial Public Library
2015 **Member**, Village of Grafton, WI Bicycle and Pedestrian Plan Committee

Journal and Proposal Reviewer

Bulletin of the American Meteorological Society
Climate Dynamics
Developmental Testbed Center
Geophysical Research Letters
Journal of Applied Meteorology and Climatology
Journal of Climate
Journal of Geophysical Research-Atmospheres
Journal of Geophysical Research-Oceans
Journal of Operational Meteorology
Journal of the Atmospheric Sciences
Monthly Weather Review
National Environment Research Council (UK)
National Oceanic and Atmospheric Administration (USA; multiple programs)
National Science Foundation (USA)
Nature Communications
Quarterly Journal of the Royal Meteorological Society
U.S. Department of Energy Atmospheric System Research Program
Weather and Forecasting

Invited Professional Colloquia and Seminars

2026 **Purdue Univ., Dept. of Earth, Atmospheric, and Planetary Sciences**
"NOAA/GSL Model Development and Forecasting Activities using MPAS for Future NOAA Operational Modeling Systems"

2024 **Univ. of Arizona, Dept. of Hydrology and Atmospheric Sciences**
"Real-Time High-Resolution Hurricane Prediction with the Model for Prediction Across Scales"

2024 **NSF National Center for Atmospheric Research, Research Applications Laboratory**
"Real-Time High-Resolution Hurricane Prediction with the Model for Prediction Across Scales"

2024 **North Carolina State Univ., Dept. of Marine, Earth, and Atmospheric Sciences**
"High-Impact Tropical and Midlatitude Weather Phenomena Across Scales" and
"Moving MEAS Forward, Together: A Vision for Leadership and for the Dept. of Marine, Earth, and Atmospheric Sciences"

2023 **NOAA Global Systems Laboratory**

- “Diagnosing Atmospheric Boundary Layer Analysis and Forecast Biases in Short-Range Numerical Weather Prediction Forecasts”
- 2023 National Science Foundation**
“Perspectives on the CLD and PDM Programs and Diversity, Equity, and Inclusion”
- 2022 6th Midwest Student Conference on Atmospheric Research**
“The Extratropical Transition of Tropical Cyclones (and Assorted Career Musings)”
- 2019 IOGP Metocean Committee**
“Tropical Cyclone Impacts at Higher Latitudes in a Warming World”
- 2018 NOAA/NWS/Storm Prediction Center**
“A Preliminary Evaluation of Paired Regional/Convection-Allowing Model-Forecast Vertical Profiles in Warm-Season, Thunderstorm-Supporting Environments”
- 2018 Northern Illinois Univ., Dept. of Geography**
“The Rear-Inflow Jet Evolution of Idealized, Mature Mesoscale Convective Systems”
- 2018 Greater Milwaukee Chapter of the AMS**
“The Harvey-Irma-Maria Hurricanes: An Atlantic Hurricane Season Retrospective”
- 2017 St. Cloud State Univ., Dept. of Atmospheric and Hydrologic Sciences**
“The Rear-Inflow Jet Evolution of Idealized, Mature Mesoscale Convective Systems”
- 2016 Lyndon State College, Dept. of Atmospheric Sciences**
“Understanding Trends in and Controls on Atlantic Hurricane Season Length”
- 2016 Univ. of Wisconsin-Madison, Dept. of Atmospheric and Oceanic Sciences**
“On the Short- to Medium-range Predictability of Thunderstorm Formation”
- 2015 Greater Milwaukee Chapter of the AMS**
“How do Forecasters Utilize Ensembles? Case Study of a High-Impact Event”
- 2014 Central Michigan Univ., Dept. of Earth and Atmospheric Sciences**
“The Predictability of Mesoscale Convective Phenomena”
- 2014 Omaha/Offutt Chapter of the AMS/NWA**
“How do Forecasters Utilize Output from a Convection-Permitting Ensemble Forecast System? Case Study of a High-Impact Precipitation Event”
- 2014 Univ. of Georgia, Dept. of Geography**
“Oklahoma's Tropical Storm: The Curious Case of T.S. Erin's Inland Reintensification”
- 2013 Greater Milwaukee Chapter of the AMS**
“Anatomy of a Superstorm: Birth, Evolution, and Impacts of Hurricane Sandy (2012)”

2012	UW-Milwaukee, Atmospheric Science Club Fall: "The 8 May 2009 'Super Derecho': A High-Impact Convective Event" Spring: "A Primer on Numerical Weather Prediction and Ensemble Modeling"
2011	Florida State Univ., Dept. of Earth, Ocean, and Atmospheric Science "A Unique Pathway to Tropical Cyclogenesis: Tropical Storm Erin (2007)"
2010	UW-Milwaukee, Dept. of Mathematical Sciences "A Unique Pathway to Tropical Cyclogenesis: Tropical Storm Erin (2007)"
2009	NCAR, Mesoscale and Microscale Meteorology Division "The Thermodynamic Evolution of Recurving Tropical Cyclones"
2007	Bermuda Institute of Ocean Sciences, RPI Research Update "Development of Anomalous Probability Forecasts for the Threat of Higher Latitude Hurricane Impacts"

Invited Workshops and Meteorological Testbeds

2026	NOAA Numerical Weather Prediction Moonshot Workshop NOAA National Center for Weather and Climate Prediction, College Park, MD
2x, last: 2026	NOAA Winter Weather Experiment NOAA/NWS/NCEP/Weather Prediction Center, College Park, MD
2025	NOAA Flash Flood and Intense Rainfall Experiment NOAA/NWS/NCEP/Weather Prediction Center, College Park, MD
13x, last: 2026	NOAA Hazardous Weather Testbed Spring Forecasting Experiment NOAA/OAR/NSSL and NOAA/NWS/NCEP/SPC, Norman, OK
2024	Innovations in Open Science Planning Workshop: Community Expectations for a Geoscience Data Commons NSF National Center for Atmospheric Research, Boulder, CO
2023	NCAR-NOAA Community Modeling Infrastructure Meeting UCAR/NSF NCAR, Boulder, CO
2022	Mind the Gap 2 Workshop Natl. Science Foundation, Amer. Meteor. Society, and Univ. at Albany, Albany, NY
2022, 2020	EarthCube Research Coordination Network "What About Model Data?" Workshops Univ. of North Dakota, Grand Forks, ND and UCAR, Boulder, CO
2012	"Shaping the Development of EarthCube to Enable Advances in Data Assimilation and Ensemble Prediction" Workshop Unidata/National Science Foundation, Boulder, CO
2006	"The Challenge of Convective Forecasting" Summer Colloquium UCAR/Advanced Study Program, Boulder, CO

Public Interviews, Presentations, and Web Stories

2026 **Earth Prediction Innovation Center** ([March 2026](#); MPAS selection for NOAA ops)
 2025 **Global Systems Laboratory** ([25 Sept 2025](#); NOAA testbed evaluations of GSL tools)
 2025 **Global Systems Laboratory** ([3 July 2025](#); story on GSL modeling advances)
 2024 **DTC Transitions Newsletter** ([Spring 2024](#); Director's Corner article on AI modeling)
 2024 **Ozaukee News Graphic** ([11 April 2024](#); weather monitoring station in Grafton, WI)
 2023 **USA Today** ([20 October 2023](#); hurricane model interpretation)
 2023 **Learning in Retirement Waukesha County** (13 September 2023; Wisconsin weather)
 2023 **Grafton, WI Public Library** (9 September 2023; observing Wisconsin weather)
 2023 **Brookfield, WI Public Library** (6 September 2023; observing Wisconsin weather)
 2023 **Wisconsin Examiner** ([5 September 2023](#); August 2023 heat + climate adaptation)
 2023 **WUWM Public Radio** ([13 June 2023](#); Wisconsin drought conditions)
 2023 **WDJT-TV/CBS 58** ([16 March 2023](#); El Niño and Wisconsin weather)
 2023 **WUWM Public Radio** ([3 February 2023](#); warm January weather + climate change)
 2023 **Milwaukee Journal Sentinel** ([19 January 2023](#); warm January weather)
 2023 **Milwaukee Journal Sentinel** ([5 January 2023](#); early-winter temperatures and snow)
 2022 **UWM Report** ([10 November 2022](#); hurricane data resource)
 2022 **Learning in Retirement Waukesha County** (17 October 2022; hurricane primer)
 2022 **WISN-TV** (7 September 2022; "Rooftop Weather" interview)
 2022 **Wall Street Journal** ([16 August 2022](#); early-starting Atlantic hurricane seasons)
 2022 **UWM Osher Lifelong Learning Institute** (21 February 2022; hurricane primer)
 2021 **WUWM Public Radio** ([5 November 2021](#); summer weather and climate change)
 2021 **350.org Milwaukee Chapter** ([14 September 2021](#); The Climate Consensus)
 2021 **WUWM Public Radio** ([4 August 2021](#); derechos and wildfires)
 2021 **UWM Research Magazine** ([16 March 2021](#); hurricane research)
 2021 **UWM Today** ([18 February 2021](#); hurricane research)
 2021 **UWM Alumni Association Master Chat** ([18 February 2021](#); hurricane primer)
 2020 **WDJT-TV/CBS 58** ([13 December 2020](#); La Niña and Wisconsin winter weather)
 2020 **WDJT-TV/CBS 58** ([14 June 2020](#); extreme rainfall in Wisconsin)
 2020 **UWM Alumni Magazine** ([8 June 2020](#); Spring 2020 semester reflection)
 2020 **Song-a-Day #4045** ([28 January 2020](#); songification of Evans et al. (2014) abstract)
 2019 **AMS On The Air Podcast** ([18 June 2019](#); extratropical transition of tropical cyclones)
 2019 **Developmental Testbed Center Newsletter** ([Spring 2019](#); vision for NWP research)
 2019 **UWM Report** ([2 May 2019](#); weather balloon launch)
 2019 **WISN-TV** (1 May 2019; weather balloon launch)
 2019 **Ozaukee News Graphic** (14 March 2019; letter to the editor on weather vs. climate)
 2019 **Ozaukee Press** (6 March 2019; letter to the editor on weather vs. climate)
 2019 **DTC Transitions Newsletter** ([Spring 2019](#); Director's Corner article on the FV3 model)
 2018 **UWM Letters & Science InFocus** ([October 2018](#); Science-A-Thon participation)
 2018 **WDJT-TV/CBS 58** (14 September 2018; Hurricane Florence interview)
 2018 **Weather Underground** ([6 June 2018](#); Tropical Storm Alberto interview)
 2018 **UWM Report** ([19 April 2018](#); Atmospheric Science program changes)
 2018 **UWM Today** ([19 April 2018](#); Innovative Weather and Atmospheric Science program)
 2018 **UWM Report** ([18 April 2018](#); late-ending winter weather in Wisconsin)
 2018 **Weather Underground** ([19 February 2018](#); Tropical Cyclone Kelvin interview)
 2017 **WISN-TV** (30 August 2017; Hurricane Harvey interview)
 2017 **Milwaukee Area Science Advocates** ([13 July 2017](#); "Actual Living Scientist" sketch)
 2017 **UWM Atmospheric Science Promo** ([February 2017](#); undergraduate program)
 2016 **News@Unidata** ([23 May 2016](#); deployment of AWIPS II at UWM)
 2016 **UWM PantherVision** (4 March 2016; El Niño impacts on Wisconsin weather)
 2016 **The Daily Beast** ([15 January 2016](#); Hurricane Alex interview)
 2015 **WITI-TV/FOX 6** ([25 February 2015](#); Milwaukee Air & Water Show weather)
 2015 **WITI-TV/FOX 6** ([24 February 2015](#); Lake Michigan ice cover)

2015	WDJT-TV/CBS 58 (23 February 2015; Lake Michigan ice cover)
2014	UWM Report (20 November 2014 ; UWM StormReady University designation)
2012	UCAR News (2 July 2012 ; eastern United States derecho)
2011	UCAR News (6 May 2011 ; Tropical Storm Erin research)
2010	LiveScience (12 September 2010 ; central United States derecho)
2010	UCAR News (18 August 2010 ; central United States derecho)

Presentations

(*underlined italics denotes student advisee*)

2026

- Barkas, K., and **C. Evans**, 2026: Quantifying the overlap between severe-weather and hurricane seasons in the southeast United States. *Abstract, 21st Symp. on Societal Applications*, Houston, TX, Amer. Meteor. Soc., 11.4.
- Bernardet, L., and Coauthors, 2026: Integrating MPAS into NOAA. *Abstract, 2026 MPAS/WRF Users' Workshop*, Boulder, CO, NSF NCAR, TBD.
- Clark, A., and Coauthors, 2026: MPAS evaluations for severe weather forecasting during the 2026 NOAA/Hazardous Weather Testbed Spring Forecasting Experiment. *Abstract, 2026 MPAS/WRF Users' Workshop*, Boulder, CO, NSF NCAR, TBD.
- Dowell, D., and Coauthors, 2026: Spring 2026 severe weather forecasting with experimental CONUS 3-km MPAS systems. *Abstract, 2026 MPAS/WRF Users' Workshop*, Boulder, CO, NSF NCAR, TBD.
- Evans, C.**, S. Trahan, C. R. Holt, and L. Bernardet, 2026: Real-time global variable-resolution hurricane forecasts using the Model for Prediction Across Scales during the 2025 Hurricane Forecast Improvement Program. *Abstract, 16th Conf. on Transition of Research to Operations*, Houston, TX, Amer. Meteor. Soc., 14.2.
- Evans, C.**, S. Trahan, C. R. Holt, B. W. Green, and L. Bernardet, 2026: Real-time global variable-resolution hurricane forecasts using the Model for Prediction Across Scales during the 2025 Hurricane Forecast Improvement Program Real-time Experiment. *Abstract, 2026 MPAS/WRF Users' Workshop*, Boulder, CO, NSF NCAR, TBD.
- Evans, C.**, S. Trahan, C. R. Holt, B. W. Green, and L. Bernardet, 2026: Real-time global variable-resolution hurricane forecasts using the Model for Prediction Across Scales during the 2025 and 2026 Hurricane Forecast Improvement Program. *Abstract, 30th Conf. on Numerical Weather Prediction*, Madison, WI, Amer. Meteor. Soc., TBD.
- Evans, C.**, C. R. Alexander, L. Bernardet, T. T. Ladwig, M. Hu, D. Dowell, and T. I. Alcott, 2026: NOAA/GSL model development and forecasting activities toward RRFsv2 using MPAS. *Abstract, 16th Conf. on Transition of Research to Operations*, Houston, TX, Amer. Meteor. Soc., J4.1.
- Evans, C.**, C. R. Alexander, L. Bernardet, T. T. Ladwig, M. Hu, D. Dowell, R. Ahmadov, and G. Ge, 2026: NOAA/GSL model development and forecasting activities using MPAS for future NOAA operational modeling systems. *Abstract, 2026 MPAS/WRF Users' Workshop*, Boulder, CO, NSF NCAR, TBD.
- Evans, C.**, C. R. Alexander, L. Bernardet, T. T. Ladwig, M. Hu, D. Dowell, R. Ahmadov, and G. Ge, 2026: NOAA/GSL model development and forecasting activities using MPAS for future NOAA operational modeling systems. *Abstract, 2026 Unifying Innovations in Forecasting Capabilities Workshop*, College Park, MD, Earth Prediction Innovation Center, TBD.
- Evans, C.**, C. R. Alexander, L. Bernardet, T. T. Ladwig, M. Hu, D. Dowell, R. Ahmadov, and G. Ge, 2026: NOAA/GSL model development and forecasting activities using MPAS for future NOAA operational modeling systems. *Abstract, 30th Conf. on Numerical Weather Prediction*, Madison, WI, Amer. Meteor. Soc., TBD.
- Ge, G., and Coauthors, 2026: Developing the prototype MPAS- and JEDI-based Atmospheric River Prediction System. *Abstract, 2026 MPAS/WRF Users' Workshop*, Boulder, CO, NSF NCAR, TBD.

- Jensen, A. A., G. Ge, M. Barlage, **C. Evans**, and J. B. Olson, 2026: An evaluation of TEMPO microphysics in MPAS for atmospheric river prediction. *Abstract, 30th Conf. on Numerical Weather Prediction*, Madison, WI, Amer. Meteor. Soc., TBD.
- Kavulich, Jr., M., K. Newman, M. Harrold, **C. Evans**, and D. Dowell, 2026: Evaluating MPAS-based RRFs prototype forecasts using METplus. *Abstract, 30th Conf. on Numerical Weather Prediction*, Madison, WI, Amer. Meteor. Soc., TBD.
- Ladwig, T. T., and Coauthors, 2026: Rapid Refresh Forecast System (RRFS) research to support aviation hazards forecasts. *Abstract, 26th Conf. on Aviation, Range, and Aerospace Meteorology*, Houston, TX, Amer. Meteor. Soc., J9.4.
- Li, H., and Coauthors, 2026: Improving the aerosol representation and microphysics coupling in NOAA's RRFsv2 numerical weather prediction model with MPAS dynamical core. *Abstract, 2026 MPAS/WRF Users' Workshop*, Boulder, CO, NSF NCAR, TBD.
- Li, H., and Coauthors, 2026: Improving the aerosol representation and microphysics coupling in NOAA's RRFsv2 numerical weather prediction model with MPAS dynamical core. *Abstract, 2026 Unifying Innovations in Forecasting Capabilities Workshop*, College Park, MD, Earth Prediction Innovation Center, TBD.
- Olson, J. B., X. Sun, B. Adler, L. Bianco, A. Jensen, D. D. Turner, **C. Evans**, J. Sedlar, and K. Balmes, 2026: HRRR/RRFS physics development for improved marine clouds in WFIP3. *Abstract, 30th Conf. on Numerical Weather Prediction*, Madison, WI, Amer. Meteor. Soc., TBD.
- Smith, M., **C. Evans**, N. Babij, R. Pierce, I. McGinnis, G. Padmanabhan, and J. A. Hamilton, 2026: Predictability of rapid intensification in the 2024 and 2025 Atlantic hurricane seasons: using METplus to verify model performance. *Abstract, 16th Conf. on Transition of Research to Operations*, Houston, TX, Amer. Meteor. Soc., 15A.3.
- Stuber, D. A., **C. Evans**, I. Jankov, and D. S. Abdi, 2026: Implicit learning of surface physics in data-driven models: an evaluation using 2-m temperature. *Abstract, 25th Conf. on Artificial Intelligence for Env. Sci.*, Houston, TX, Amer. Meteor. Soc., 8A.5.
- Sun, X., J. B. Olson, B. Adler, L. Bianco, **C. Evans**, and D. D. Turner, 2026: Characterizing sea-breeze events using observations and model simulations during WFIP3. *Abstract, 30th Conf. on Numerical Weather Prediction*, Madison, WI, Amer. Meteor. Soc., TBD.
- Sun, X., J. Olson, L. Bianco, B. Adler, D. Turner, J. Sedlar, A. Jensen, and **C. Evans**, 2026: A comparison of MPAS and WRF-ARW forecasts of coastal/marine case studies during WFIP3. *Abstract, 2026 MPAS/WRF Users' Workshop*, Boulder, CO, NSF NCAR, TBD.
- Toy, M. D., **C. Evans**, J. Beck, W. Skamarock, and S. Bhimireddy, 2026: Preventing computational instabilities and spurious precipitation features in the lateral boundaries of high-resolution regional MPAS. *Abstract, 2026 MPAS/WRF Users' Workshop*, Boulder, CO, NSF NCAR, TBD.
- Trahan, S., **C. Evans**, and C. Holt, 2026: Global variable resolution hurricane model: troubles and revelations. *Abstract, 2026 MPAS/WRF Users' Workshop*, Boulder, CO, NSF NCAR, TBD.

2025

- Adams-Selin, R. D., H. Vagasky, D. V. Blount, and **C. Evans**, 2025: Low-frequency gravity waves within mesoscale convective system stratiform regions. *Abstract, 4th Symp. on Mesoscale Processes*, New Orleans, LA, Amer. Meteor. Soc., 9.5.
- Adams-Selin, R. D., H. Vagasky, D. V. Blount, and **C. Evans**, 2025: Low-frequency gravity waves within mesoscale convective system stratiform regions. *Abstract, 2025 ARM/ASR Joint User Facility and PI Meeting*, Rockville, MD, US Dept. of Energy, 3.45.
- Barkas, K. R., and **C. Evans**, 2025: Toward quantifying the overlap between severe weather and hurricane seasons in the southeast United States. *Abstract, 24th Student Conf.*, New Orleans, LA, Amer. Meteor. Soc., S31.
- Blount, D. V., **C. Evans**, and R. D. Adams-Selin, 2025: A preliminary analysis of the contributions of line-end vortices, gravity waves, and environmental flow to mesoscale convection system rear inflow and stratiform region structure in numerical simulations. *Abstract, 4th Symp. on Mesoscale Processes*, New Orleans, LA, Amer. Meteor. Soc., 9.4.
- Blount, D. V., **C. Evans**, R. D. Adams-Selin, and H. Vagasky, 2025: An analysis of the contributions of line-end vortices and gravity waves to mesoscale convective system rear inflow and

- stratiform region structure in numerical simulations. *Abstract, 2025 ARM/ASR Joint User Facility and PI Meeting*, Rockville, MD, US Dept. of Energy, 3.47.
- Boyle, K. G., C. Evans, and S. Kravtsov, 2025: Downstream effects of Northern Hemisphere extratropical transition in a future climate. *Abstract, 38th Conf. on Climate Variability and Change*, New Orleans, LA, Amer. Meteor. Soc., 897.
- Clark, A., and coauthors, 2025: MPAS evaluations for severe weather forecasting during the 2025 NOAA/Hazardous Weather Testbed Spring Forecasting Experiment. *Abstract, Joint MPAS/WRF Users Workshop 2025*, Boulder, CO, Natl. Cen. for Atmos. Res., 4.5.
- Evans, C.**, 2025: Performance of a real-time, high-resolution Model for Prediction Across Scales (MPAS) forecast system during HFIP 2024. *Abstract, 2025 Tropical Cyclone Operations and Research Forum*, Lakeland, FL, Natl. Oceanic and Atmos. Administration, 3.4.
- Evans, C.**, C. R. Alexander, L. R. Bernardet, T. T. Ladwig, M. Hu, D. C. Dowell, and T. I. Alcott, 2025: NOAA/GSL model development and forecasting activities using MPAS. *Abstract, Joint MPAS/WRF Users Workshop 2025*, Boulder, CO, Natl. Cen. for Atmos. Res., 3.1.
- Evans, C.**, C. R. Alexander, L. R. Bernardet, T. T. Ladwig, M. Hu, D. C. Dowell, and T. I. Alcott, 2025: NOAA/GSL model development and forecasting activities using MPAS. *Abstract, Unifying Innovations in Forecasting Capabilities Workshop*, Boulder, CO, Earth Prediction Innovation Center.
- Evans, C.**, L. R. Bernardet, M. D. Toy, J. B. Olson, A. A. Jensen, W. C. Skamarock, and M. J. Duda, 2025: Integration of additional community physics parameterizations into the MPAS-Atmosphere model. *Abstract, Joint MPAS/WRF Users Workshop 2025*, Boulder, CO, Natl. Cen. for Atmos. Res., P27.
- Evans, C.**, L. Bengtsson, A. Jensen, S. Trahan, N. Wang, M. Barlage, G. Grell, and E. Grell, 2025: Modeling system advancements for atmospheric river applications. *NOAA Atmospheric River Workshop*, virtual, Natl. Oceanic and Atmos. Administration, 5.1.
- Hamilton, J., M. Smith, R. Pierce, N. Babij, I. McGinnis, G. Padmanabhan, M. Wandishin, **C. Evans**, D. Turner, and S. Weygandt, 2025: Use of MATS (Model Assessment Tool Suite) for rapid-prototyping model refinement. *NOAA Atmospheric River Workshop*, virtual, Natl. Oceanic and Atmos. Administration, 6.2.
- Jensen, A. A., J. B. Olson, and **C. Evans**, 2025: Development updates to the Thompson-Eidhammer Microphysics Parameterization for Operations (TEMPO). *Abstract, Joint MPAS/WRF Users Workshop 2025*, Boulder, CO, Natl. Cen. for Atmos. Res., 6B.1.
- Li, H., R. Ahmadov, A. A. Jensen, J. Schnell, J. B. Olson, M. Zhang, and **C. Evans**, 2025: A pilot study of improving the aerosol representation in NOAA's RRFsv2 numerical weather prediction model with the MPAS dynamical core. *Abstract, 2025 AGU Fall Meeting*, New Orleans, LA, Amer. Geophys. Union, A31H - Poster.
- Olson, J. B., W. M. Angevine, X. Sun, D. Turner, and **C. Evans**, 2025: Updates to the MYNN-EDMF PBL scheme to improve operational forecasting applications. *Abstract, Joint MPAS/WRF Users Workshop 2025*, Boulder, CO, Natl. Cen. for Atmos. Res., 6A.1.

2024

- Blount, D. V., C. Evans, R. D. Adams-Selin, and H. Vagasky, 2024: A preliminary analysis of line-end vortex contributions to rear-to-front flow in observed and modeled mesoscale convective systems. *Abstract, 24th Symp. on Meteor. Obs. and Instrumentation*, Baltimore, MD, Amer. Meteor. Soc.
- Blount, D. V., C. Evans, and R. D. Adams-Selin, 2024: A preliminary analysis of the contributions of line-end vortices, gravity waves, and environmental flow to mesoscale convection system rear inflow and stratiform region structure in numerical simulations. *Abstract, 31st Conf. on Severe Local Storms*, Virginia Beach, VA, Amer. Meteor. Soc.
- DeYoung, C. P., and **C. Evans**, 2024: A preliminary assessment of the HRRR's ability to predict the Great Lakes lake-breeze front and marine atmospheric boundary layer. *Abstract, 22nd Symp. on the Coastal Environment*, Baltimore, MD, Amer. Meteor. Soc., 1.3.
- Evans, C.**, 2024: Tropical cyclone forecasts at GSL using MPAS. *Abstract, 2024 HFIP Annual Meeting*, Miami, FL, Natl. Oceanic and Atmos. Administration, 3C.4.

- Hickok, A. O., M. P. Vossen, and C. Evans, 2024: Towards an updated climatology of overland tropical cyclone maintenance and intensification in non-/weakly baroclinic environments. *Abstract, 6th Spec. Symp. on Tropical Meteor. and Trop. Cyclones*, Baltimore, MD, Amer. Meteor. Soc.
- Hickok, A. O., M. P. Vossen, and C. Evans, 2024: Towards an updated climatology of overland tropical cyclone maintenance and intensification in non-/weakly baroclinic environments. *Abstract, 16th UWM Undergraduate Research Symposium*, Milwaukee, WI.
- Hickok, A. O., M. P. Vossen, and C. Evans, 2024: Towards an updated climatology of overland tropical cyclone maintenance and intensification in non-/weakly baroclinic environments. *Abstract, 36th Conf. on Hurricanes and Tropical Meteorology*, Long Beach, CA, Amer. Meteor. Soc.
- Vossen, M. P., and C. Evans, 2024: Thermodynamics of overland tropical cyclone intensity change in weakly/non-baroclinic environments. *Abstract, 36th Conf. on Hurricanes and Tropical Meteorology*, Long Beach, CA, Amer. Meteor. Soc.

2023

- Adams-Selin, R. D., **C. Evans**, H. C. Vagasky, and D. V. Blount, 2023: Identification of low-frequency gravity waves within mesoscale convective system stratiform regions. *Abstract, 2023 Joint ARM User Facility and ASR PI Meeting*, Rockville, MD, US Dept. of Energy, 4.48.
- Blount, D. V., C. Evans, and R. D. Adams-Selin, 2023: A preliminary analysis of low-frequency gravity wave, line-end vortex, and environmental flow contributions to rear-to-front flow in observed MCSs. *Abstract, 3rd Symp. on Mesoscale Processes*, Denver, CO, Amer. Meteor. Soc., 283.
- Blount, D. V., C. Evans, R. D. Adams-Selin, and H. Vagasky, 2023: A preliminary analysis of low-frequency gravity wave, line-end vortex, and environmental flow contributions to rear-to-front flow in observed MCSs. *Abstract, 20th Conf. on Mesoscale Processes*, Madison, WI, Amer. Meteor. Soc., 7.2.
- Brown, G. R. H., and coauthors, 2023: The Climate Consensus network – empowering current and future scientists to engage in climate outreach within our universities. *Abstract, 32nd Conf. on Education*, Denver, CO, Amer. Meteor. Soc., 74.
- DeYoung, C. P., and C. Evans, 2023: A preliminary assessment of the HRRR's ability to predict the Great Lakes lake-breeze front and marine atmospheric boundary layer. *Abstract, 2023 Great Lakes Operational Meteorology Workshop*, Madison, WI.
- DeYoung, C. P., and C. Evans, 2023: A preliminary assessment of the High-Resolution Rapid Refresh model's ability to predict the Great Lakes lake-breeze front and marine atmospheric boundary layer. *Abstract, 32nd Conf. on Weather Analysis and Forecasting*, Madison, WI, Amer. Meteor. Soc., 86.
- Hickok, A. O., M. P. Vossen, and C. Evans, 2023: Towards an updated climatology of overland tropical cyclone maintenance and intensification in non-/weakly baroclinic environments. *Abstract, 22nd Annual Student Conference*, Denver, CO, Amer. Meteor. Soc., S253.
- Hickok, A. O., M. P. Vossen, and C. Evans, 2023: Towards an updated climatology of overland tropical cyclone maintenance and intensification in non-/weakly baroclinic environments. *Abstract, 15th UWM Undergraduate Research Symposium*, Milwaukee, WI.
- Hickok, A. O., M. P. Vossen, and C. Evans, 2023: Towards an updated climatology of overland tropical cyclone maintenance and intensification in non-/weakly baroclinic environments. *Abstract, 32nd Conf. on Weather Analysis and Forecasting*, Madison, WI, Amer. Meteor. Soc., 106.
- Prince, K. C., and C. Evans, 2023: Convectively generated negative potential vorticity enhancing the jet stream through an inverse energy cascade during the extratropical transition of Hurricane Irma. *Abstract, 5th Spec. Symp. on Tropical Meteorology and Tropical Cyclones*, Denver, CO, Amer. Meteor. Soc., 14.4.
- Vagasky, H., R. D. Adams-Selin, **C. Evans**, and D. V. Blount, 2023: Observations of convectively generated gravity waves within the stratiform region of mesoscale convective systems. *Abstract, 20th Conf. on Mesoscale Processes*, Madison, WI, Amer. Meteor. Soc., 32.

Vossen, M. P., and **C. Evans**, 2023: Thermodynamics of overland tropical cyclone intensity change in weakly/non-baroclinic environments. *Abstract, 5th Spec. Symp. on Tropical Meteorology and Tropical Cyclones*, Denver, CO, Amer. Meteor. Soc., 8.4.

2022

Adams-Selin, R. D., J. Mascio, and **C. Evans**, 2022: Establishing a holistic understanding of the circulations of mesoscale convective system stratiform regions. *Abstract, 2022 Joint ARM/ASR PI Meeting*, Rockville, MD, 4.58.

Blount, D. V., **C. Evans**, I. L. Jirak, A. Dean, and S. Kravtsov, 2022: An objective vertical thermodynamic profile shape classification method: formulation and application to forecast verification. *Abstract, 31st Conf. on Weather Analysis and Forecasting/27th Conf. on Numerical Weather Prediction*, Houston, TX, Amer. Meteor. Soc., J14.1.

Hanrahan, J., and coauthors, 2022: Building capacity for climate change outreach: supporting, encouraging, and inspiring scientists within our academic institutions. *Abstract, 2022 Earth Educators Rendezvous*, Minneapolis, MN, Natl. Assoc. of Geoscience Teachers.

Hanrahan, J., and coauthors, 2022: Creating a multi-institution outreach network to improve climate literacy. *Abstract, 10th Symp. on the Weather, Water, and Climate Enterprise*, Houston, TX, Amer. Meteor. Soc., 274.

Hanrahan, J., and coauthors, 2022: The Climate Consensus network – creating capacity for climate outreach within our universities. *Abstract, AGU Fall Meeting*, Chicago, IL, ED15C-0376.

Metz, N. D., and coauthors, 2022: Atmospheric rivers over the Northeast United States. *Abstract, AGU Fall Meeting*, Chicago, IL, A55M-1275.

Prince, K. C., **C. Evans**, and S. Kravtsov, 2022: A case-study analysis of convective-scale contributions to tropical cyclones' interactions with the midlatitude waveguide. *Abstract, 35th Conf. on Hurricanes and Tropical Meteorology*, New Orleans, LA, Amer. Meteor. Soc., 13C.3.

Prince, K. C., **C. Evans**, and S. Kravtsov, 2022: A case-study analysis of convective-scale contributions to tropical cyclones' interactions with the midlatitude waveguide. *Abstract, MeteoXchange ECS Conference*, Virtual/Online, Germany Federal Ministry of Education and Research, 5.1.

Prince, K., and **C. Evans**, 2022: The importance of convective-scale processes in a recent tropical cyclone-midlatitude waveguide interaction. *Abstract, 31st Conf. on Weather Analysis and Forecasting/27th Conf. on Numerical Weather Prediction*, Houston, TX, Amer. Meteor. Soc., J9.1.

Spencer, M. R., and **C. Evans**, 2022: The influence of mesoscale sea-surface temperature uncertainty on short-range forecasts of cold-season southeast United States severe weather events. *Abstract, 19th Conf. on Mesoscale Processes*, Houston, TX, Amer. Meteor. Soc., 11.5.

Spencer, M. R., and **C. Evans**, 2022: The influence of mesoscale sea-surface temperature uncertainty on short-range forecasts of cold-season southeast United States severe weather events. *Abstract, 30th Conf. on Severe Local Storms*, Santa Fe, NM, Amer. Meteor. Soc., 407141.

Vossen, M. P., and **C. Evans**, 2022: An investigation of thermodynamic maintenance and intensification mechanisms of tropical cyclones over land. *Abstract, 31st Conf. on Weather Analysis and Forecasting/27th Conf. on Numerical Weather Prediction*, Houston, TX, Amer. Meteor. Soc., 1A.5.

Vossen, M. P., and **C. Evans**, 2022: Thermodynamics of overland tropical cyclone intensity change in weakly/non-baroclinic environments. *Abstract, 35th Conf. on Hurricanes and Tropical Meteorology*, New Orleans, LA, Amer. Meteor. Soc., 14B.5.

2021

Blount, D. V., **C. Evans**, I. L. Jirak, and A. Dean, 2021: Verifying GFS short-range-forecast vertical thermodynamic profiles using an objective profile-shape classification method. *Abstract,*

- 11th Conf. on Transition of Research to Operations, New Orleans, LA, Amer. Meteor. Soc., 5A.7.
- Kaminski, A. N., and **C. Evans**, 2021: Toward a satellite-based cyclone classification routine: a modern 3-yr climatology of North Atlantic and western North Pacific extratropical cyclones. *Abstract, 20th Student Conference*, New Orleans, LA, Amer. Meteor. Soc., 26.
- Kaminski, A. N., and **C. Evans**, 2021: A modern 3-year climatology of North Atlantic and Western North Pacific extratropical cyclones. *Abstract, 13th UWM Undergraduate Research Symposium*, Milwaukee, WI.
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2019

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2018

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Evans, C., 2006: Dynamics of the wind field expansion with extratropically transitioning tropical cyclones. *Extended Abstract, 27th Conf. on Hurricanes and Tropical Meteorology*, Monterey, CA, Amer. Meteor. Soc., 4A.6.

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Hart, R. E., J. L. Evans, and **C. Evans**, 2005: Synoptic composites of the ET lifecycle of North Atlantic TCs: Factors determining post-transition evolution. *Proceedings, 3rd Intl. Workshop on Extratropical Transition*, Perth, Australia.

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Professional Memberships & Honor Societies

2010-2012	American Geophysical Union
2005	Chi Epsilon Pi, Florida State University Chapter
2004	Phi Beta Kappa, Alpha Chapter of Florida
2003	National Society of Collegiate Scholars
2002-present	American Meteorological Society